

Scenario 1: Identify Consistent High Performers

Question: Find employees who have consistently received a salary increase for the past 3 years.

Steps:

1. SELECT the employee's salary records for each of the past 3 years from the Salary (or EmployeeSalary) table, including year information.
2. Use GROUP BY employee to organize salaries year-wise for each employee.
3. Use the HAVING clause (and possibly window functions or self-joins) to check that the salary for each consecutive year is strictly greater than the previous year (WHERE salary_year2 > salary_year1 AND salary_year3 > salary_year2).

Scenario 2: Customer Retention Analysis

Question: Find customers who made purchases in the last 6 months but not in the previous 6 months.

Steps:

1. SELECT customer IDs from the Orders/Sales table where the purchase date is within the last 6 months (WHERE purchase_date BETWEEN [today minus 6 months] AND [today]).
2. SELECT customer IDs who made purchases in the previous 6 months (WHERE purchase_date BETWEEN [today minus 12 months] AND [today minus 6 months]).
3. Use the EXCEPT or NOT IN keyword to filter customers who appear only in the last 6 months and not in the previous 6 months.

Scenario 3: Identify Products with Declining Sales

Question: Find products whose sales decreased by more than 30% compared to the previous month.

Steps:

1. SELECT the product ID (or name), and the total sales amount for both the current month and the previous month from the sales table (using WHERE clause to filter by month or GROUP BY product and month).
2. Use a JOIN or window function to pair each product's current and previous month sales totals.

3. Use a WHERE clause to calculate the percentage decrease: $(\text{current_month_sales} < \text{previous_month_sales} \times 0.7)$ or $(\text{previous_month_sales} - \text{current_month_sales}) / \text{previous_month_sales} > 0.3$.

Scenario 4: Calculate Moving Average of Sales

Question: Calculate a 3-month moving average for product sales.

Steps:

1. SELECT product ID (or name), sales amount, and month from the sales table.
2. Use a window function like `AVG(sales_amount) OVER (PARTITION BY product_id ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)` to calculate the 3-month moving average for each product.
3. Include the moving average as a column in the SELECT statement.

Scenario 5: Detect Duplicate Records

Question: Find all duplicate records from a customer table based on email and phone number.

Steps:

1. SELECT the email and phone number columns from the Customer table.
2. Use the GROUP BY clause on email and phone number to group records with the same values.
3. Use the HAVING clause to filter groups that have a count greater than 1 (`HAVING COUNT(*) > 1`), which indicates duplicates.

Scenario 6: Identify Products with No Sales for 3 Consecutive Months

Question: List products that have not had any sales for 3 consecutive months.

Steps:

1. SELECT product ID (or name) and month from the sales table, showing months with sales records for each product.
2. Find all products and months (could use a calendar or generate all month-product combinations).
3. Use a window function (LAG or COUNT) or self-join to check for 3 consecutive months with no sales records (missing months).